

Set 1

Standard: 8

Subject: Science

Time: 1 Hour

Max Marks: 20

Chapter 16: Life Cycle of Stars

Q.1. Objectives (4 Marks)

A. Fill in the blanks.

1. The process of energy generation in the Sun is due to the fusion of _____ nuclei.
2. The distance travelled by light in one year is called a _____.

B. State whether the following statements are True or False.

1. A star is born from huge clouds of gas and dust called interstellar clouds.
2. The mass of the Sun is about 3.3 lakh times that of the earth.

Q.2. Short Answer (Pick 5, Solve any 4) (8 Marks)

1. What are the different constituents of our solar system?
2. What are the major differences between a star and a planet?
3. Which is the star nearest to us?
4. What is a galaxy?
5. What is a light year?

Q.3. Brief/Long Answer (Pick 3, Solve any 2) (8 Marks)

1. How do stars remain stable?
2. Explain the birth of stars.
3. Why do stars evolve?

Set 2

Standard: 8

Subject: Science

Time: 1 Hour

Max Marks: 20

Chapter 16: Life Cycle of Stars

Q.1. Objectives (4 Marks)

A. Find the odd man out.

1. Spiral galaxy, Elliptical galaxy, Irregular galaxy, Solar system.
2. White dwarf, Neutron star, Black hole, Planet.

B. Fill in the blanks.

1. Our galaxy is a _____ galaxy and is called the Milky Way.
2. The life cycle of a star depends on its initial _____.

Q.2. Short Answer (Pick 5, Solve any 4) (8 Marks)

1. What is the reason to call Sun an ordinary star?
2. What happens when a gas sphere contracts?
3. What are the three types of galaxies?
4. How is the mass of other stars measured?
5. What is the unit used by scientists for measuring large distances in the universe?

Q.3. Brief/Long Answer (Pick 3, Solve any 2) (8 Marks)

1. Explain the three end stages of stars based on their mass.
2. What will happen to the Sun in about 4.5 billion years?
3. What are the properties of stars?

ANSWER KEYS

Set 1

Q.1. Objectives

A. 1. Hydrogen, 2. Light year

B. 1. True, 2. True

Q.2. Short Answers

1. Constituents of Solar System:

- The Sun is at the center of the solar system.
- It consists of eight planets orbiting the Sun.
- There are satellites of various planets.
- It also contains asteroids and comets.

2. Star vs. Planet:

- Stars are self-luminous objects that emit their own light.
- Planets are non-luminous and reflect light from stars.
- Stars are made of hot gases and have very high temperatures.
- Planets are much smaller than stars and have lower temperatures.

3. Nearest Star:

- The star nearest to the Earth is the Sun.
- The star closest to the Sun is alpha Centauri.
- Light takes about 8 minutes to reach us from the Sun.
- Light takes about 4.2 years to reach us from alpha Centauri.

4. Galaxy:

- A galaxy is a collection of billions of stars.
- It includes their planetary systems and interstellar clouds.
- There are empty spaces between the stars in a galaxy.
- The universe is made up of innumerable such galaxies.

5. Light Year:

- A light year is a unit of distance used in astronomy.
- It is the distance travelled by light in one year.
- The speed of light is 3,00,000 km/s.
- One light year is equal to 9.5×10^{12} km.

Q.3. Brief/Long Answers

1. Stability of Stars:

- The stability of a star is maintained by a balance between two opposing forces.
- Gravitational force acts inwards, trying to bring gas particles together.
- Gas pressure, created by the hot gas, acts outwards from the center.
- When these two forces are balanced, the star remains stable.
- If gravitational force exceeds gas pressure, the star contracts.
- If gas pressure exceeds gravitational force, the star expands.

2. Birth of Stars:

- Huge interstellar clouds of gas and dust exist in galaxies.
- Due to some disturbance, these clouds start contracting.
- Contraction increases the density and temperature of the cloud.
- A dense sphere of hot gas is formed from the cloud.
- Once temperature and density increase sufficiently, nuclear fusion starts.
- The gas sphere becomes self-luminous, and a star is born.

3. Evolution of Stars:

- Evolution means a change in a star's properties over time.
- Stars pass through different stages during their lifetime.
- The reason for evolution is the burning and decrease of fuel at the center.
- When fuel finishes, energy generation stops and temperature decreases.
- Decreased temperature leads to decreased gas pressure, causing contraction.
- Contraction triggers the burning of another fuel, repeating the cycle.

Set 2

Q.1. Objectives

A. 1. Solar system, 2. Planet

B. 1. Spiral, 2. Mass

Q.2. Short Answers

1. Sun as an Ordinary Star:

- It appears larger because it is much closer to Earth.
- There are billions of stars with masses and sizes similar to the Sun.
- Some stars are much smaller or larger than the Sun.
- Others are much hotter or cooler than the Sun.

2. Gas Sphere Contraction:

- The temperature of the gas sphere increases during contraction.
- This happens due to the transformation of gravitational potential energy.
- The potential energy is converted into heat energy.
- Higher density also leads to higher gas pressure.

3. Types of Galaxies:

- Galaxies are classified based on their structure and shape.
- Spiral galaxies have a disc-like shape with a bulge.
- Elliptical galaxies are shaped like an ellipse.
- Irregular galaxies do not have a defined shape.

4. Measuring Mass of Stars:

- Masses of stars are very large and difficult to measure in kg.
- They are measured relative to the mass of the Sun.
- The mass of the Sun (M_{Sun}) is used as the standard unit.
- Other stars typically have masses between 0.1 to 100 M_{Sun} .

5. Distance Unit:

- Large distances in the universe are measured in light years.

- A light year is the distance light travels in one year.
- Light takes 8 minutes from the Sun and 4.2 years from alpha Centauri.
- Sizes of interstellar clouds are also expressed in light years.

Q.3. Brief/Long Answers

1. End Stages of Stars:

- The end stage of a star depends on its initial mass.
- Stars with mass less than $8 M_{\text{Sun}}$ end as white dwarfs.
- In a white dwarf, electron pressure balances gravity forever.
- Stars with mass between 8 and $25 M_{\text{Sun}}$ end as neutron stars.
- In neutron stars, pressure from neutrons balances gravitational force.
- Stars with mass greater than $25 M_{\text{Sun}}$ end as black holes.

2. Future of the Sun:

- The Sun has remained unchanged for about 4.5 billion years.
- It will continue to stay stable for another 4.5 billion years.
- After that, its properties will slowly begin to change.
- It will eventually evolve into a red giant star.
- Its diameter will increase enough to swallow Mercury and Venus.
- The Earth may also be absorbed by the Sun in this stage.

3. Properties of Stars:

- Stars have distinct physical properties like mass and radius.
- Their radius can range from 0.1 to 1000 times that of the Sun.
- Their mass can range from 0.1 to 100 times that of the Sun.
- Surface temperature varies greatly among different stars.
- The age of stars ranges from a few million to billions of years.
- Most stars are mainly composed of hydrogen and helium.